Paul Perez

Aerospace Engineer in Planetary Sciences

Having worked in the fields of atmospheric modelling and emergency response scenarios has shown me some of the key environmental and societal challenges we face today. And applying scientific innovations to solve issues and preserve our planet is what drives me. Which is why I now look forward to further tackling these challenges through the lens of Earth monitoring and remote sensing, contributing to a collective of like-minded individuals.

CONTACT ME

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Relocating to Barcelona in 2026

przpaul.github.io

EXPERTISE

- Atmospheric Modelling
- Climate Data Handling
- Scientific Programming (Julia / Python / Unix shell)

CERTIFICATION

AI for Earth Monitoring **(EUMETSAT)**

INTERESTS

- Climate Change Mitigation
- Open-ended Discussions
- Outdoor Rock Climbing

LANGUAGES

French
Native Proficiency

English
Native Proficiency

Spanish
Intermediate Proficiency

WORK EXPERIENCE

Scientific Researcher in Dispersion Modelling — 01/2024 - 08/2025

Royal Military Academy of Belgium (RMA) I Belgium

- Developed web application for CBRN particle dispersion modelling, based on ECMWF meteorological data and computational infrastructure.
- Implemented NATO procedure and Lagrangian dispersion model FLEXPART.
- Integrated probabilistic prediction approach with weather forecast ensembles.
- Constructed dispersion and radiation case study for enhanced emergency preparedness.

Intern in WRF-GHG Modelling — 06/2020 - 10/2020

Netherlands Institute for Space Research (SRON) I Netherlands

- Established ensemble of Weather Research and Forecasting (WRF) model physics configurations suitable for high resolution CO_2 transport modelling based on literature.
- Quantified uncertainties in CO_2 mole fraction due to transport errors in WRF model.
- Identified main physics options and signals contributing to the uncertainties present.

EDUCATION

MSc Aerospace Engineering – Space Flight Track — 2019 - 2023

Delft University of Technology (TU Delft) I Netherlands

- Modules: Climate Change Science and Ethics, Physics of the Earth and Atmosphere, Planetary Sciences, MicroSat Engineering, Numerical Astrodynamics, and more.
- Successfully defended Thesis project about the impact of interstitial air pressure on the phenomenon of booming sand acoustic emissions, in the context of Mars exploration and extra-terrestrial soil characterisation, using modelling and laboratory experiments.

BEng Mechanical, Materials and Aerospace Engineering — 2016 - 2019

INSA Lyon I France

- Modules: Fluid Mechanics, FEM, Engineering Mathematics, Aerodynamics, Propulsion Systems, Experiments and Simulations, Materials Science, and more.
- Successfully developed and delivered graphical user interface for MECALAM's power transmission simulation software using PyQt software, for final year duo project.

PUBLICATION

Perez, P., Benhassine, M., Quinn, J., Janssens, B., & Van Utterbeeck, F. (2025). Simulation of a nuclear fallout scenario in the Zaporizhzhia nuclear power plant during the Ukraine conflict: Implications of dose rate distributions for downwind populations. *Military Medicine, 190*(Supplement_2), 536–542. https://doi.org/10.1093/milmed/usaf263

REFERENCES

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